Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

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Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

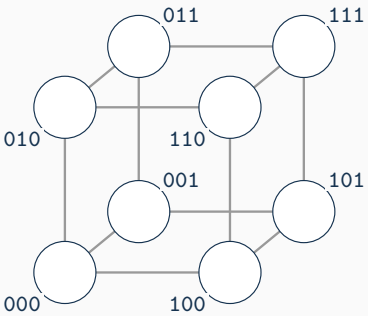
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arXiv:2610.09341

Outline

1. **CSS codes and transversal gates**
2. The $[\![8, 3, 2]\!]$ code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Generator rows on the cube

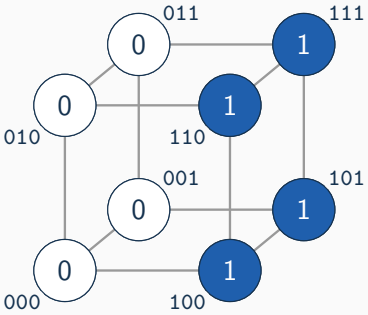


One qubit at each point $u \in \mathbb{F}_2^3$.

Each column is a point $u \in \mathbb{F}_2^3$ (a qubit), each row a Boolean function f, with entry $f(u)$. Weight: $|f| = \#\{u : f(u) = 1\}$.

$f \setminus u$	000	001	010	011	100	101	110	111

Generator rows on the cube

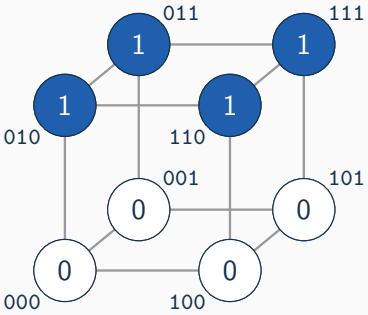


Row u_1: the face $u_1 = 1$.

Each column is a point $u \in \mathbb{F}_2^3$ (a qubit), each row a Boolean function f, with entry $f(u)$. Weight: $|f| = \#\{u : f(u) = 1\}$.

$f \setminus u$	000	001	010	011	100	101	110	111
u_1	0	0	0	0	1	1	1	1

Generator rows on the cube

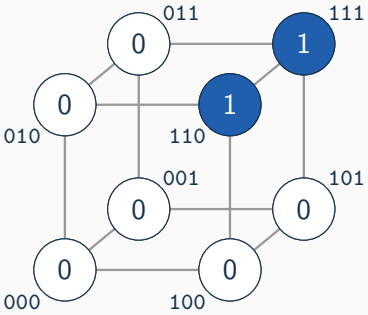


Row u_2: the face $u_2 = 1$.

Each column is a point $u \in \mathbb{F}_2^3$ (a qubit), each row a Boolean function f, with entry $f(u)$. Weight: $|f| = \#\{u : f(u) = 1\}$.

$f \setminus u$	000	001	010	011	100	101	110	111
u_1	0	0	0	0	1	1	1	1
u_2	0	0	1	1	0	0	1	1

Generator rows on the cube



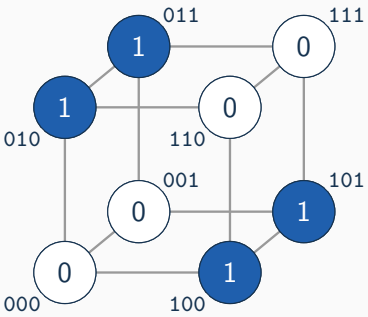
u_1u_2: an edge.

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u_1u_2	0	0	0	0	0	0	1	1

Multiplying functions overlaps their rows: $c \wedge c'$ (entrywise product). Dot product: $c \cdot c' = |c \wedge c'| \bmod 2$.

Generator rows on the cube



$u_1 + u_2$.

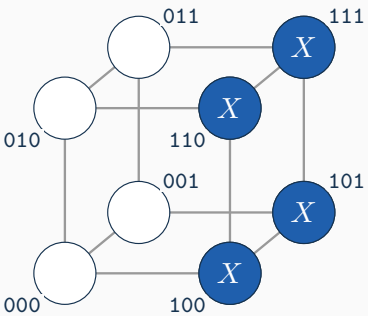
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$f \setminus u$	000	001	010	011	100	101	110	111
u_1	0	0	0	0	1	1	1	1
u_2	0	0	1	1	0	0	1	1
$u_1 u_2$	0	0	0	0	0	0	1	1
$u_1 + u_2$	0	0	1	1	1	1	0	0

Multiplying functions overlaps their rows: $c \wedge c'$ (entrywise product). Dot product: $c \cdot c' = |c \wedge c'| \bmod 2$.

Adding functions XORs their rows: $c \oplus c'$ (entrywise sum).

Pauli strings

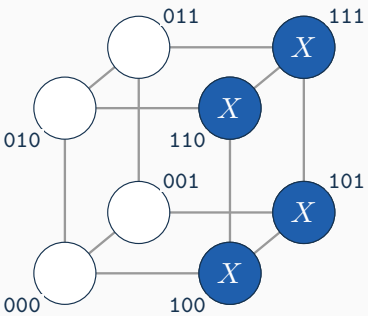


$X(u_1)$: X on the face $u_1 = 1$.

Pauli strings $X(a) = \bigotimes_j X^{a_j}$ and $Z(b) = \bigotimes_j Z^{b_j}$, for $a, b \in \mathbb{F}_2^n$, act on basis states $|c\rangle$:

$$X(a)|c\rangle = |c \oplus a\rangle, \qquad Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle.$$

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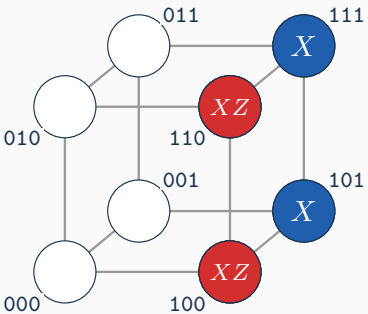
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They commute or anticommute:

$$X(a)Z(b) = (-1)^{a\cdot b} Z(b)X(a).$$

Pauli strings



Red: X and Z. Overlap 2.

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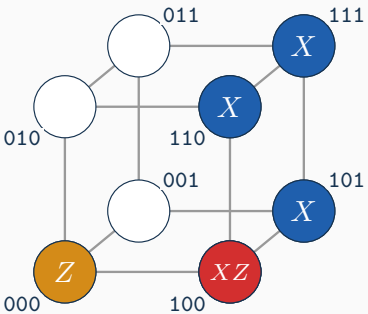
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With $a = u_1$ and $\delta_u \in \mathbb{F}_2^8$ the weight-one vector at u:

$$b = \delta_{100} \oplus \delta_{110}: \quad a \cdot b = 0, \text{ commute,}$$

Pauli strings



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$$b = \delta_{100} \oplus \delta_{110}: \quad a \cdot b = 0, \text{ commute},$$
$$b = \delta_{000} \oplus \delta_{100}: \quad a \cdot b = 1, \text{ anticommute}.$$

Stabilizer and CSS codes

Stabilizer group $\mathcal{S}$: abelian, generated by Pauli strings (the *stabilizer generators*), $-I \notin \mathcal{S}$. The code is the subspace fixed by $\mathcal{S}$. *Syndrome*: the measured signs of the generators. An error flips the signs of those it anticommutes with.

Logical operator: a Pauli string commuting with $\mathcal{S}$, nontrivial if not in $\mathcal{S}$. $[\![n, k, d]\!]$: n qubits, k logical qubits, d the least weight of a nontrivial logical operator.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
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CSS code of $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96], with s stabilizer rows S_l and k logical rows K_i, has X-type and Z-type stabilizers

$$X(a),\ a \in \mathcal{C}_2 = \operatorname{rowspan} S, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}, \qquad \mathcal{C}_1 = \operatorname{rowspan} G, \quad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

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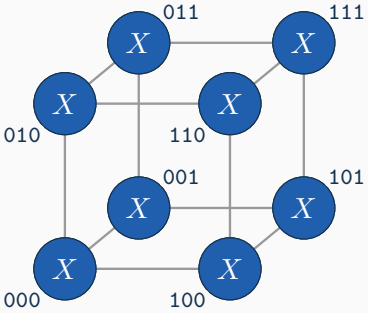
$$X(a),\ a \in \mathcal{C}_2 = \operatorname{rowspan} S, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}, \qquad \mathcal{C}_1 = \operatorname{rowspan} G, \quad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

Logical basis: uniform superposition over the coset $K^\top x \oplus \mathcal{C}_2$,

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad K^\top x = \bigoplus_i x_i K_i.$$

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The $[\![8, 3, 2]\!]$ code



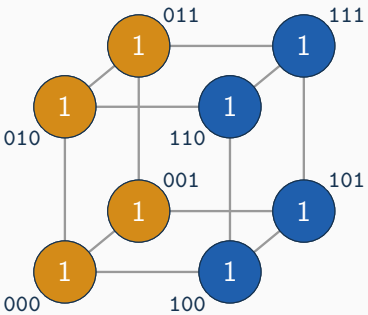
$X(\mathbf{1})$, the X-type stabilizer generator.

$G = [K; S]$:

	000	001	010	011	100	101	110	111
$K_1 = u_1$	0	0	0	0	1	1	1	1
$K_2 = u_2$	0	0	1	1	0	0	1	1
$K_3 = u_3$	0	1	0	1	0	1	0	1
$S = \mathbf{1}$	1	1	1	1	1	1	1	1

$\mathcal{C}_1 = \mathrm{RM}(1, 3)$, the functions of degree ≤ 1: an $[8, 4, 4]$ code.

The $[\![8, 3, 2]\!]$ code



$x = 100$. Blue: $K_1 = 00001111$. Gold: 11110000.

$G = [K; S]$:

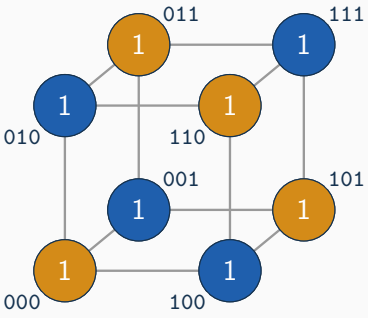
	000	001	010	011	100	101	110	111
$K_1 = u_1$	0	0	0	0	1	1	1	1
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With $K^\top x = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$|x\rangle_L = \big(|K^\top x\rangle + |\mathbf{1} \oplus K^\top x\rangle\big)/\sqrt{2}.$$

The $[\![8, 3, 2]\!]$ code



$x = 111$. Blue: $K_1 \oplus K_2 \oplus K_3 = 01101001$. Gold: its complement.

$G = [K; S]$:

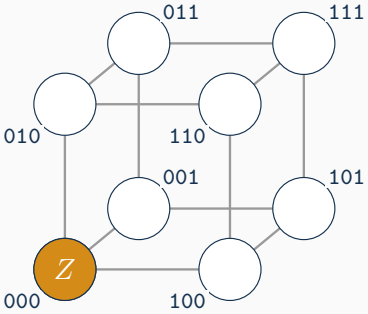
	000	001	010	011	100	101	110	111
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Z on 000: syndrome 1.

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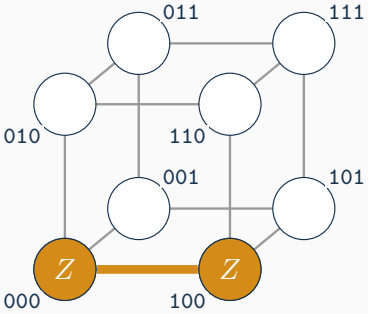
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$Z(\delta_{000})$ anticommutes with $X(\mathbf{1})$: syndrome 1.

The $[\![8, 3, 2]\!]$ code



Z on the edge 000–100: syndrome 0.

$G = [K; S]$:

	000	001	010	011	100	101	110	111
$K_1 = u_1$	0	0	0	0	1	1	1	1
$K_2 = u_2$	0	0	1	1	0	0	1	1
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$Z(\delta_{000} \oplus \delta_{100})|x\rangle_L = (-1)^{x_1}|x\rangle_L$: a logical operator.

Nontrivial X-, Z-type logical operators have least weights $d_X = 4$, $d_Z = 2$, so $d = 2$.

Clifford and non-Clifford gates

On basis states, with $x, x_i \in \mathbb{F}_2$:

$$H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$
$$T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*, and $|+\rangle = H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

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Clifford gates alone are not universal. Adding T or CCZ makes them universal.

Transversal gates and distillation

A physical unitary $\bar{U}$ implements the logical gate U if it maps every encoded state $\sum_x \alpha_x |x\rangle_L$ to the encoding of $U \sum_x \alpha_x |x\rangle$. It is *transversal* if

$$\bar{U} = U_1 \otimes \cdots \otimes U_n, \qquad U_j \text{ acting on qubit } j.$$

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
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Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal transversal gate set.

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Magic-state distillation [BK05] uses Clifford operations:

$$\text{many noisy } T|+\rangle \text{ or } \mathrm{CCZ}|{+}{+}{+}\rangle \longrightarrow \text{ fewer, less noisy copies.}$$

Distillation needs a code with a transversal logical T or CCZ.

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Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

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Every CSS code has transversal CNOT, so [EK09] rules out transversal T for Steane and H for $[\![15,1,3]\!]$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

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Every CSS code has transversal CNOT, so [EK09] rules out transversal T for Steane and H for $[\![15,1,3]\!]$.

Which CSS codes with $k = 3$ have $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ logical CCZ?

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

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Signed weight and the gate condition

A $T/T^\dagger$ *pattern* puts T^{Γ_j} on qubit j, with $\Gamma \in \{\pm 1\}^n$ and the *signed weight* sw_Γ:

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

$\mathrm{sw}_\Gamma(c) =$ (number of ones of c on T qubits) $-$ (number of ones of c on $T^\dagger$ qubits).

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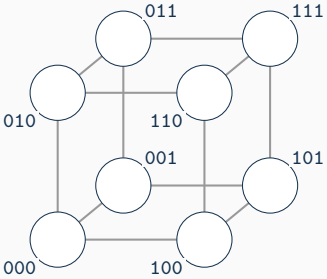
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The pattern is diagonal, so each codeword gets its own phase. Logical CCZ: every codeword of the coset $K^\top x \oplus \mathcal{C}_2$ gets the same phase $(-1)^{x_1x_2x_3}$:

$$\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

Finding a $T/T^\dagger$ pattern means choosing Γ so that this congruence holds for every codeword.

A $T/T^\dagger$ pattern for logical CCZ



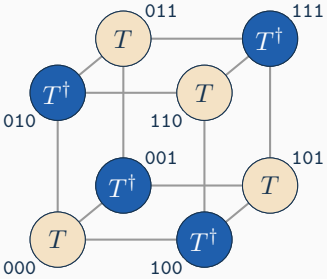
XOR of bits as an integer:

$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

Applied at each vertex to $c = K^\top x$, weighted by Γ_u:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = \sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

A $T/T^\dagger$ pattern for logical CCZ



$T^\dagger$ where $\gamma = u_1 + u_2 + u_3$ is 1 [CH17].

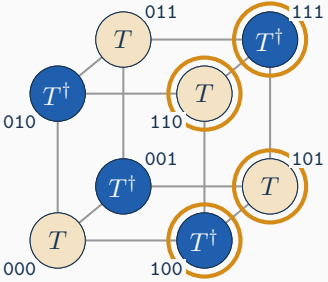
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Row $K_1 = 00001111$: the face $u_1 = 1$.

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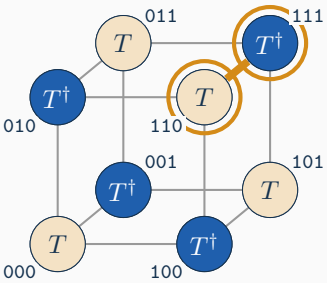
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$$\mathrm{sw}_\Gamma(K_i) = 0,$$

A $T/T^\dagger$ pattern for logical CCZ



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$K_1 \wedge K_2 = 00000011$: an edge.

XOR of bits as an integer:

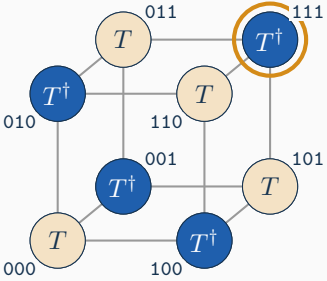
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$K_1 \wedge K_2 \wedge K_3 = 00000001$: a vertex.

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$$\mathrm{sw}_\Gamma(K_i) = 0, \quad \mathrm{sw}_\Gamma(K_i \wedge K_j) = 0, \quad \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1.$$

$\mathrm{sw}_\Gamma(c) = -4x_1x_2x_3 \equiv 4x_1x_2x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The *Bravyi–Haah parity identity* [BH12], for any binary vectors:

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1}\, \mathrm{sw}_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Comparing coefficients in the gate condition gives, for distinct rows g, g', g'' of G:

$$\begin{aligned}
\mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8,\\
\mathrm{sw}_\Gamma(g \wedge g') &\equiv 0 \pmod 4,\\
\mathrm{sw}_\Gamma(g \wedge g' \wedge g'') &\equiv \begin{cases} 1, & \{g, g', g''\} = \{K_1, K_2, K_3\},\\ 0, & \text{otherwise,} \end{cases} \pmod 2.
\end{aligned}$$

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

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The $[\![8, 3, 2]\!]$ code satisfies all three (faces and edges give 0, the vertex 111 gives -1). We impose them on the rows of G to build larger codes.

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Triorthogonal codes and CCZ

2005 Bravyi–Kitaev [BK05]: magic-state distillation.

2012 Bravyi–Haah [BH12]: triorthogonal matrices for transversal T.

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2024 [CLM+24, CLM+26]: CSS-T codes ($T^{\otimes n}$ logical), then in 2026 all diagonal transversal gates of a CSS code.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

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A *triorthogonal matrix* [BH12]: $|g \wedge g'| \equiv |g \wedge g' \wedge g''| \equiv 0 \pmod 2$ for distinct rows g, g', g''.

CCZ needs one odd triple, $K_1 \wedge K_2 \wedge K_3$, so we use the generalized conditions of [CH17] on the next slide.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

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Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

The three row conditions reduce mod 2 to the nine Campbell–Howard (CH) conditions [CH17]:

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A $T/T^\dagger$ pattern with no correction needs the mod 8 and mod 4 conditions on Γ.

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Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$\mathcal{Q}_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A $T/T^\dagger$ pattern with $26\,T$ and $22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- $\mathcal{C}_2$ is fixed by projective geometry: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$, the nonzero vectors of $\mathbb{F}_2^6$ outside a 4-dimensional subspace.
- The logical rows K come from a SAT (Boolean satisfiability) search, and the signs from linear algebra over $\mathbb{Z}_4$.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- Removing other subspaces gives codes at 13 block lengths from 48 to 496.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$.

- $\beta_j \neq 0$ for all j: every single Z error is detected.
- $\beta_i \neq \beta_j$ for all $i \neq j$: every pair of Z errors is detected.

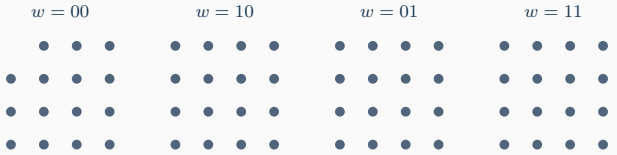
Such S is called *projective*: its columns are distinct points of $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$. In $[\![8, 3, 2]\!]$, $S = \mathbf{1}$ has all columns equal, so Z on two qubits can have syndrome 0: that is why $d_Z = 2$ there.

$d_Z \geq 3$ follows from the columns of S alone, for every choice of logical rows K.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

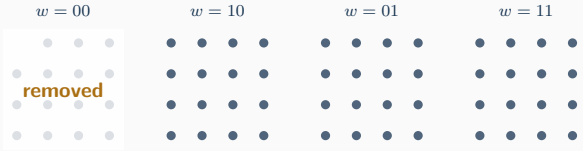
The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$, with rows indexed by (u_1, u_2) and columns by (u_3, u_4), in Gray order.



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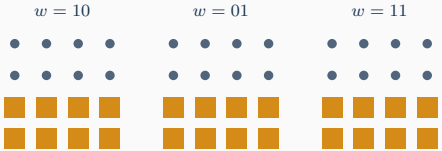


The rows of S are the 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$, evaluated on the 48 points.

Three blocks of 16 points give a projective $[48, 6]$ code $\mathcal{C}_2$.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c = \sum_i \lambda_i u_i + \sum_j \mu_j w_j$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



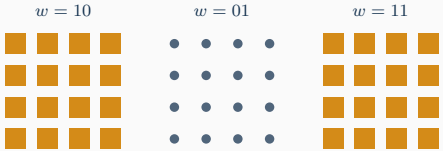
$\lambda \neq 0$, for example $c = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $c = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q-1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p\backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

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Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

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For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

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$\mathcal{Q}_{48}$: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, f' \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, $m_J = \prod_{j \in J} z_j$ and

$$\langle \chi_G, m_J \rangle = \left| \bigwedge_{j \in J} g_j \right| \bmod 2.$$

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Since $\mathrm{RM}(3, r)^{\perp} = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) satisfies these conditions [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon \delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

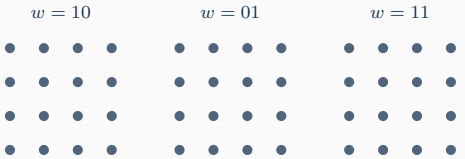
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set J of rows into its logical rows J_K and stabilizer rows J_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{J_K}(\kappa(\beta))\, m_{J_S}(\beta) = \begin{cases} 1, & J = \{K_1, K_2, K_3\}, \\ 0, & \text{else.} \end{cases}$$

S1–S3 have $J_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



Logical rows in the column picture

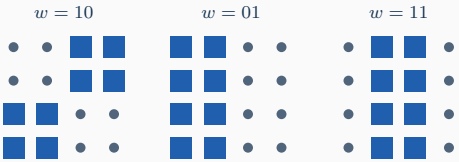
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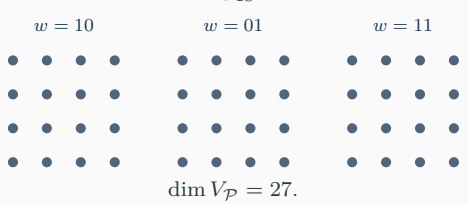
Blue: $\kappa_1 = 1$, with $\kappa_1 = u_1 + u_3$, $1 + u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Row K_i is $\kappa_i \colon \mathcal{P} \to \mathbb{F}_2$, bit i of the label map, and $\langle f, f' \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) f'(\beta) \bmod 2$ [LB26].

1. K1, KS1, KS3 say $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ (y_l: row l of S), which span $\mathrm{RM}(2, s)$:

$$\kappa_i \in V_{\mathcal{P}} = \{ f : \langle f, m \rangle_{\mathcal{P}} = 0,\ m \in \mathrm{RM}(2, s) \}.$$



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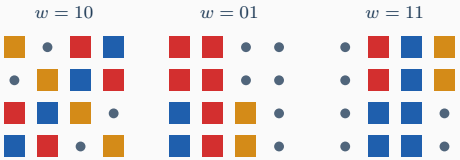
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$\mathcal{Q}_{48}$



Blue: only K_1 has a 1. Gold: only K_2 has a 1. Red: both. $|K_1 \wedge K_2| = 12 \equiv 0 \pmod 2$ (K2).

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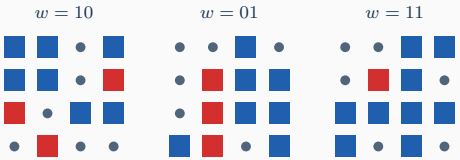
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3. The rest is linear in $\kappa_3 \in V_{\mathcal{P}}$, solved by Gaussian elimination:

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Squares: K_3 has a 1. Red: K_1 and K_2 too.
$|K_1 \wedge K_2 \wedge K_3| = 7 \equiv 1 \pmod 2$ (K3).

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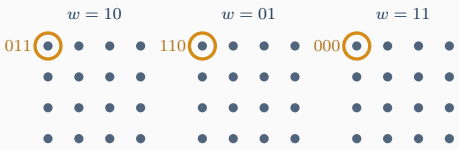
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   If no pair extends, $\mathcal{P}$ is excluded.

$\mathcal{Q}_{48}$

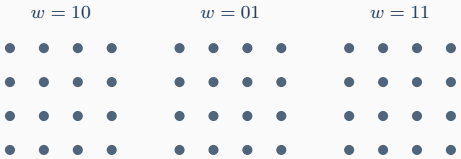


Z on the circled qubits, at $u = 0000$ (the upper left corner of each block). Their columns of S sum to 0, so no syndrome. Their labels sum to $011 \oplus 110 \oplus 000 = 101 \neq 0$, so it is logical: $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

As on the cube, $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j = 1$ on the $T^\dagger$ qubits. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$



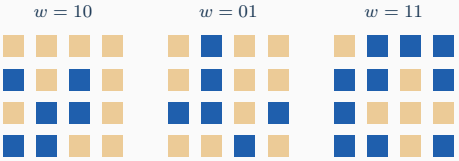
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For $\mathcal{Q}_{48}$ [LB26]:

$$\begin{aligned}\gamma(u,w) &= u_3 + w_1Q_1(u) + w_2Q_2(u),\\ Q_1 &= u_1 + u_2 + u_3 + u_1u_3 + u_2u_3 + u_2u_4,\\ Q_2 &= u_3 + u_4 + u_1u_2 + u_1u_4 + u_3u_4.\end{aligned}$$



Blue: $T^\dagger$ $(6, 6, 10$ per block). Gold: T. Total $26\,T + 22\,T^\dagger$.

Outline

1. CSS codes and transversal gates
2. The $[\![8, 3, 2]\!]$ code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. **Lower bounds and open questions**

Stabilizer supports are Reed–Muller codewords

For projective S, S1–S3 say $\langle \chi_S, m \rangle = 0$ for every monomial m in $y_1, \ldots, y_s$ of degree 1 to 3.

Following [NH22], let $c = \chi_S + \varepsilon \delta_0$, where ε makes $|c|$ even. Every monomial of degree at least 1 vanishes at 0, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = |c| \bmod 2$. Hence $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = |c| - c(0).$$

For $\mathcal{Q}_{48}$: $c = \chi_S = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce the stabilizer rank s (the number of rows of S) to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the possible weights and the geometry of the support.
3. At rank 8, use the classification of low-weight Reed–Muller codewords to reduce to finitely many supports, and enumerate them.
4. Show that no remaining support admits logical rows satisfying K1–K3 and KS1–KS3.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$, which covers quasi-transversal CCZ with $k = 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no $T/T^\dagger$ pattern [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat (a translate of a 4-dimensional subspace), with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with a circuit footprint of 9 qubits.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $\mathcal{Q}_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit logical rows K and signs Γ?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public

Backup: admissible lengths from Reed–Muller weights

S1–S3 hold if and only if $\chi_S \in \mathrm{RM}(s-4,s) \cup \big(\delta_0 + \mathrm{RM}(s-4,s)\big)$ [LB26]. At rank s the lengths are $n = |c| - c(0)$ for $c \in \mathrm{RM}(s-4,s)$ whose support spans $\mathbb{F}_2^s$.

s	code	weights ≤ 39	candidate n, $15 \leq n \leq 38$
5	$\mathrm{RM}(1,5)$	16, 32 [MS77]	15, 16, 31
6	$\mathrm{RM}(2,6)$	16, 24, 28, 32, 36 [SB70]	15, 16, 23, 24, 27, 28, 31, 32, 35, 36
7	$\mathrm{RM}(3,7)$	16, 24, 28, 32, 36 [KT70, Car21]	the same as $s = 6$

[MS77] F. J. MacWilliams and N. J. A. Sloane, The Theory of Error-Correcting Codes, North-Holland (1977).

[SB70] N. J. A. Sloane and E. R. Berlekamp, Weight enumerator for second-order Reed–Muller codes, IEEE Trans. Inf. Theory 16, 745 (1970).

[KT70] T. Kasami and N. Tokura, On the weight structure of Reed–Muller codes, IEEE Trans. Inf. Theory 16, 752 (1970).

[Car21] C. Carlet, Boolean Functions for Cryptography and Coding Theory, Cambridge University Press (2021), Sec. 4.1.5 (Ax's theorem).

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7	$\mathrm{RM}(3,7)$	16, 24, 28, 32, 36 [KT70, Car21]	the same as $s = 6$

$n = 15, 16$: then $|c| = 16$, the minimum weight, so the support is an affine 4-flat. On a flat of dimension $m \leq 4$, K1, KS1, KS3 force $\deg \kappa_i \leq m-3$, so $\deg(\kappa_1\kappa_2\kappa_3) \leq 3(m-3) \leq m-1$. Then $|K_1 \wedge K_2 \wedge K_3|$ is even, which contradicts K3.

[MS77] F. J. MacWilliams and N. J. A. Sloane, The Theory of Error-Correcting Codes, North-Holland (1977).

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Backup: lowering the stabilizer rank

Project along a nonzero $v \in \mathbb{F}_2^s$ (kernel $\{0, v\}$) and delete one row S_m with $v_m = 1$, after $K_i \mapsto K_i \oplus \tau_i S_m$. The label differences along v are

$$\Delta_v = \{\kappa(\beta \oplus v) \oplus \kappa(\beta) : \ \beta, \ \beta \oplus v \in \mathcal{P}\} \subseteq \mathbb{F}_2^3.$$

If $\Delta_v \subseteq \{\tau\}$ for some $\tau \in \mathbb{F}_2^3$, and $\kappa(v) = \tau$ when $v \in \mathcal{P}$, the result keeps the nine CH conditions, $k = 3$ and $d_Z \geq 3$ at rank $s - 1$ [LB26].

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At most $\lfloor \binom{n}{2}/2 \rfloor$ vectors v have $|\Delta_v| \geq 2$, and n lie in $\mathcal{P}$, so a good v exists when

$$2^s - 1 > n + \left\lfloor \tbinom{n}{2}/2 \right\rfloor. \qquad n \leq 38 : \ 38 + 351 = 389 < 2^9 - 1, \text{ so } s \leq 8.$$

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$s \leq 5$: of the 63 supports from $\mathrm{RM}(1, 5)$, 62 are hyperplanes or punctured hyperplanes, which are flats. The last, $\mathbb{F}_2^5 \setminus \{0\}$ with $n = 31$, fails Algorithm 1 on all $49{,}741{,}825$ pairs.

Backup: the finite cases at ranks 6, 7 and 8

s	$\lvert c\rvert$	cases	result
6, 7	16	affine 4-flat	K3 fails
6	24–36	six affine classes (Dickson forms). The degree-one class lies in a hyperplane, so the rank drops to 5	no κ_3
7	24, 28	1 and 2 classes [KT70]	no κ_3
7	32	y_1y_2 (in a hyperplane) and 3 cubic classes [KTA76]	no κ_3
7	36	9,792 functions, 7,038 supports	no κ_3
8	≤ 30	counting: $n + \lfloor \binom{n}{2}/2 \rfloor \leq 247 < 2^8 - 1$	rank drops
8	32, $\deg c \leq 3$	affine 5-flat, rank ≤ 6	excluded
8	32–38, $\deg c = 4$	$c = y_7 f + y_8 g + y_7 y_8 h$ with $\lvert c\rvert = \lvert f\rvert + \lvert g\rvert + \lvert f + g + h\rvert$, $(\lvert f\rvert, \lvert g\rvert) \in \{(8,8),(8,12),(12,12),(8,14)\}$, 2,399 orbit representatives of g, 469,728 supports	no κ_3

Every case ends in a flat or hyperplane reduction, or in Algorithm 1 finding no third logical row.

[KTA76] T. Kasami, N. Tokura, and S. Azumi, On the weight enumeration of weights less than 2.5d of Reed–Muller codes, Inf. Control 30, 380 (1976).

Backup: dictionary with [CLM+26]

[CLM+26]	This talk
$U(b) = \bigotimes_j \mathrm{diag}(1, \omega^{b_j})$, $\omega = e^{2\pi i/N}$, $N = 2^\ell$	$N = 8$, $b \equiv \Gamma$: $T = U(1)$, $T^\dagger = U(-1)$
$b \cdot w$ for $w \in \mathbb{F}_2^n$	$\mathrm{sw}_\Gamma(w)$
$U(b)$ fixes the code iff	rows with a stabilizer row: $\mathrm{sw}_\Gamma(S_l) \equiv 0$ (8),
$b \in H_N = \left(\beta_2 \star \bigcup_{i<\ell} 2^i \beta_1^{(i)}\right)^{\perp_N}$	$\mathrm{sw}_\Gamma(S_l \wedge g) \equiv 0$ (4), $\mathrm{sw}_\Gamma(S_l \wedge g \wedge g') \equiv 0$ (2)
logical action $\omega^{b\cdot(Bv)} = \prod_J C_J\text{-}U(a_J)$,	CCZ $= C_{123}\text{-}U(4)$: $\mathrm{sw}_\Gamma(K_i) \equiv 0$ (8),
$a_J = (-2)^{\lvert J\rvert-1}\, b \cdot (w_{j_1} \star \cdots \star w_{j_h})$	$\mathrm{sw}_\Gamma(K_i \wedge K_{i'}) \equiv 0$ (4),
	$\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3) \equiv 1$ (2)

Our signed conditions are their H_8 with $b \in \{\pm 1\}^n$ and the logical action fixed to CCZ. We solve the converse problem: given the action, find S, K and b.